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PERFORMANCE SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY

This specification is approved for use by U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an audio frequency amplifier (hereafter referred to as a distribution unit) used on military ground vehicles.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5820

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-462	-	Measurement of Electromagnetic Interference Characteristics, Test Method Standard for.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12310227	-	Amplifier, Audio Frequency (Interface).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASQC Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).
ANSI/SAE AS478	-	Identification Marking Methods (DoD Adopted).

(Application for copies should be addressed to American National Standard Institute, 11 West 42nd Street, New York, NY 10036.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D522	- Standard Test Methods for Mandrel Bend Test of Attached Organic Coatings (DoD Adopted).
ASTM D5487	- Standard Test Method for Simulated Drop of Loaded Containers by Shock Machines.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specification for audio amplifiers. The materials shall be capable of meeting all the operational and environmental requirements specified herein (see 4.5.1).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Fungus proof materials. To the greatest extent practical, the materials used in the distribution unit shall be non-nutrients for fungi. If materials that are nutrients for fungi must be utilized, such materials shall be approved by the Government (see 4.5.1).

3.2.3 Potting compounds. Potting compounds employed in the distribution unit shall meet all of the environmental requirements specified herein without adversely affecting the performance of the distribution unit (see 4.5.1).

3.2.4 Insulating compounds. The electrical coating for printed circuit assemblies shall meet all of the environmental requirements specified herein without adversely affecting the performance of the distribution unit (see 4.5.1).

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3.2.5 Metal parts. All ferrous metal parts shall be of corrosion resistant type or treated to provide corrosion protection and coating adherence equal to or exceeding that provided by hot dip galvanized 1010 steel with a minimum coating thickness of 50 micrometers (2 mils) on steel sheet less than 1.6 millimeters (mm) [0.0625 inches (in.)] thick, 64 micrometers (2.5 mils) on steel sheet greater than 1.6 mm (0.0625 in.) thick or 19 micrometers (0.75 mils) on pregalvanized steel sheet 1.6 mm (0.0625 in.) thick or less. A material sample shall be capable of meeting or exceeding the corrosion resistance provided by a galvanized sample (as described above) when subjected to the mandrel bend test of ASTM D522. Suitable protection against dissimilar metals and galvanic corrosion shall be provided. Magnesium shall not be used unless specifically authorized (see 4.5.1).

3.2.6 Non-metallic parts. Non-metallic parts shall be designed to minimize deterioration caused by exposure to sunlight, sand particles, microorganisms, moisture, temperature extremes, fuel, hydraulic and lubricating oil, grease, and salt degradation associated with abrasion, load, or exposure-induced deterioration that can endanger or jeopardize the function of the system (see 4.5.1).

3.3 Design and construction. The distribution unit shall be fabricated and assembled in accordance with the manufacturer's design and construction for distribution units, except for envelope dimensions, mounting dimensions and locations, and electrical mounting interfaces as shown in Drawing 12310227 and as specified herein (see 4.4.2, 4.5.1, and 4.5.2).

3.3.1 Size and weight. The distribution unit shall be 24.8 centimeters (cm) (9.75 in.) long by 14.6 cm (5.75 in.) wide by 9.5 cm (3.75 in.) deep. The weight of the distribution unit shall not exceed 3.6 kilograms (kg) [8 pounds (lbs)] (see 4.5.2 and 4.5.3).

3.3.2 Controls and interfaces.

3.3.2.1 Interface description. The distribution unit shall interface with a variety of audio accessories, radio sets (receive-transmit), receivers (radio receive only), and field wire line. Primary interface details shall be as specified in figure 1. When radio sets are used with the intercom, they are connected to the distribution unit through J501, J503, J509 and J511 receptacles. All four radio sets may be of the receive-transmit type. The retransmission function requires that two radio sets be connected to receptacles J501 and J503. This function serves as a radio relay to increase the range between these two external radio sets. When radio sets (receive-transmit) are not used, direct current (dc) power for the intercom is supplied through receptacle J508. The radio output of an auxiliary receiver may be connected to the audio input binding posts to enable monitoring. This monitoring cannot be disabled by intercom controls. Field wire lines may be connected from a local battery telephone to the line binding posts to enable two-way communication between any crew member within reach of the distribution unit's LINE/PRESS TO TALK pushbutton switch. Monitoring field wire line communications is automatic and cannot be disabled by intercom controls (see 4.5.3).

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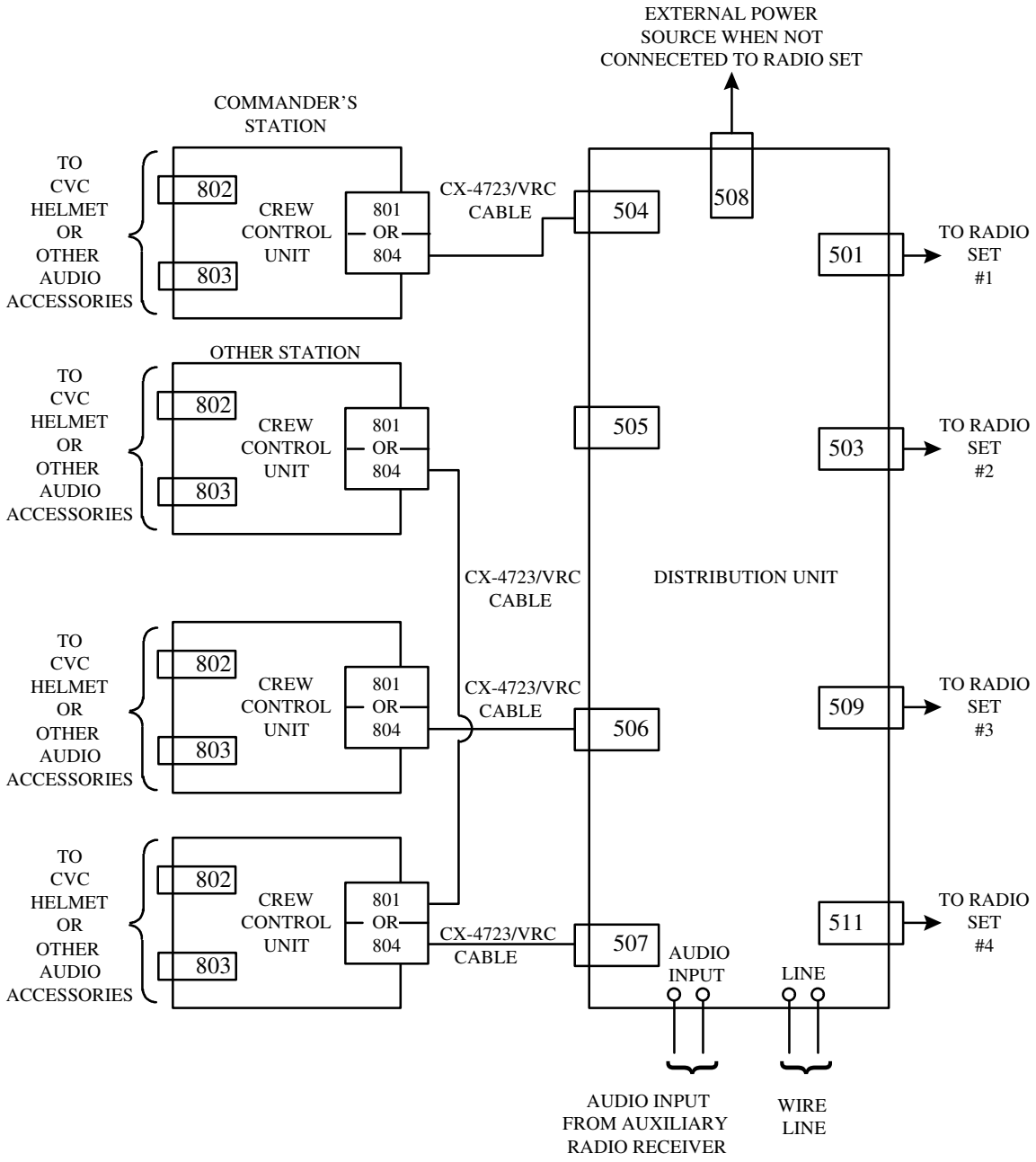


FIGURE 1. Distribution unit interface connections.

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3.3.2.2 On-off switch. A circuit breaker switch shall be provided to control power to the intercom system. This switch shall control power from all sources (see 4.5.3).

3.3.2.3 Line jacks. A pair of binding posts shall be provided to permit connection of a standard field telephone line (see 4.5.3).

3.3.2.4 Auxiliary input. A pair of binding posts shall be provided as an auxiliary input to the intercom channel (see 4.5.3).

3.3.2.5 Intercom cable assembly. A "standard" special purpose cable assembly (CX-4723/VRC) of the proper length shall be used to interconnect the distribution unit and crew control units. The cables shall be terminated by an eighteen-terminal male connector at both ends. Pin connections for this cable shall be as specified in figure 2 (see 4.5.3).

3.3.2.6 Power indicator. A green indicator lamp shall be provided to indicate when power is on. When the lens cap is rotated one quarter turn counterclockwise, the light shall not shine through the lens (see 4.5.3).

3.3.2.7 Intercom accentuated switch. When in the ON position, this switch shall reduce the audio level from all radio receivers (see 4.5.3).

3.3.2.8 Radio transmission selector switch. This switch shall control the transmitter keying lines of the radios as follows (see 4.5.3):

a. Commander and Crew (Position 1). Any crew control unit may initiate transmissions on any radio. If two or more crew members select a single radio, all may transmit on that radio.
b. Commander Only (Position 2). Only the crew control unit or units connected to J504 of the distribution unit may initiate radio transmission.
c. Listening Silence (Position 3). No crew control unit shall be permitted to initiate radio transmissions.

3.3.2.9 Radio retransmission switch (see 4.5.3).

a. In NORMAL position, all radios operate independently.
b. In RETRANSMISSION position, receive audio, transmit audio, and keying lines of the radios connected to J501 and J503 of the distribution unit shall be interconnected such that signals received by one radio are automatically retransmitted by the other radio. The connections shall be configured such that any crew control unit may monitor the audio passing between the radios, but no crew control unit may initiate transmission on either radio.

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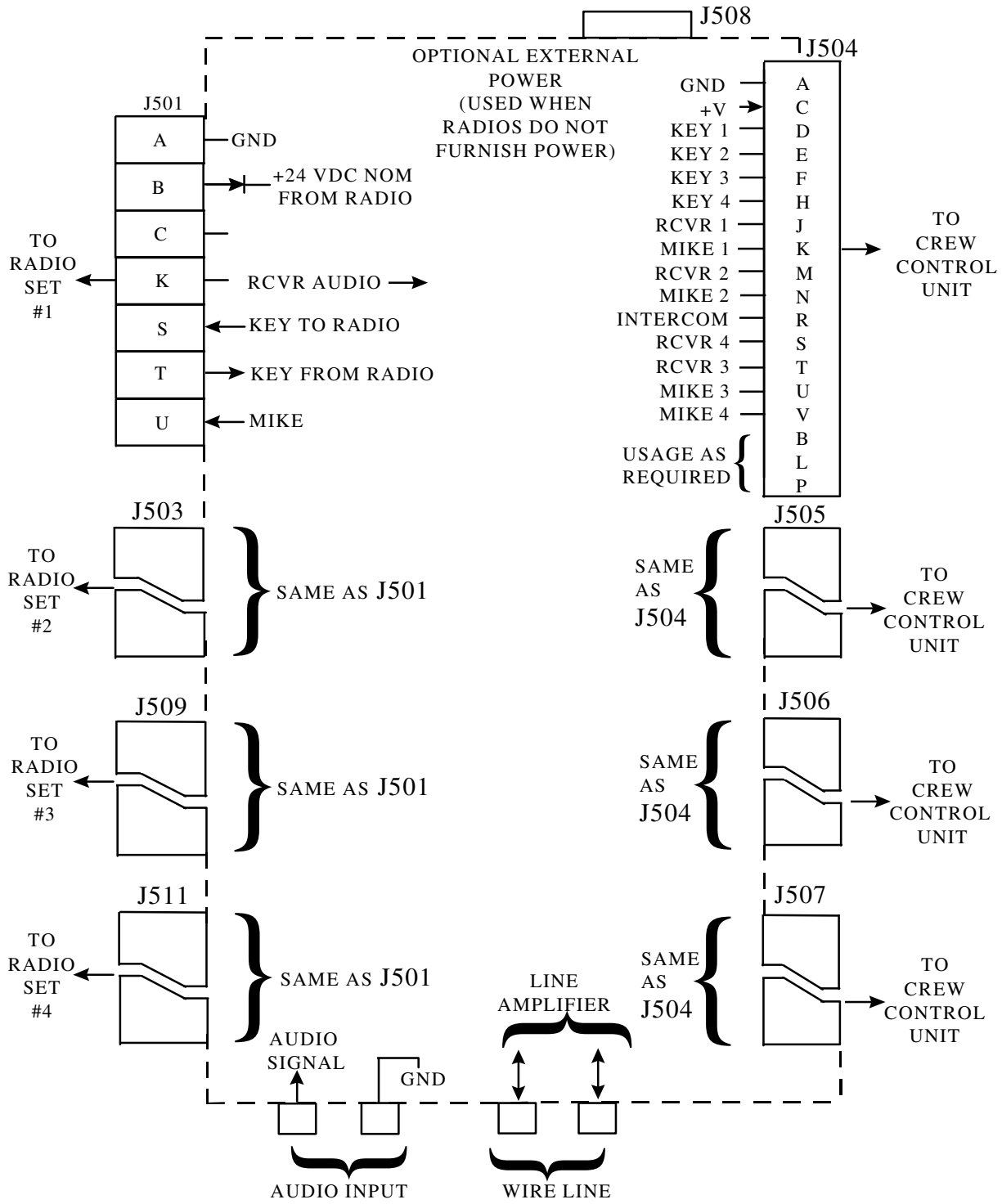


FIGURE 2. Distribution unit electrical interface.

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3.3.2.10 Line switch. Audio from a wire telephone line connected to the line terminals shall be monitored continuously by the intercom channel when the line switch is in the LISTEN position. When the switch is placed in the TALK position, audio from the intercom channel shall be transmitted to the telephone line. The switch shall be a spring loaded, momentary type, biased to the LISTEN position (see 4.5.3).

3.3.2.11 Power interface. Direct current power shall be applied to the distribution unit by any one of the connected radio sets when the radio is turned on, via pin B of J501, J503, J509, or J511. An auxiliary power input, J508, shall also be provided to permit use of the intercom without radios. All power inputs shall be diode isolated to prevent cross coupling between sources. After a radio is turned on, power-up of the intercom is achieved by setting the power circuit breaker on the distribution unit to the ON position (see 4.5.3).

3.3.2.12 Power distribution. The distribution unit shall accept supply voltage from pin B of J501, J503, J509, J511 or J508. The source shall be diode isolated to prevent cross coupling of power between connectors (see 4.5.3).

3.3.2.13 Retransmission switch. When the retransmission switch is placed in the RETRANSMISSION position, pin T of J501 shall be connected to pin S of J503, and pin S of J501 shall be connected to pin T of J503. Likewise, receiver audio from pin K of J501 shall be directed to pin U of J503, and receiver audio from pin K of J503 shall be directed to pin U of J501. Receiver audio shall be available for monitoring by crew members at any crew control unit by selecting the appropriate monitor switches; however, it shall not be possible for any crew control unit to initiate a radio transmission on the radios connected to J501 or J503 when in the RETRANSMISSION mode (see 4.5.3).

3.3.3 Interface parts. All interface parts shall be in accordance with Drawing 12310227 (see 4.5.1).

3.4 Performance.

3.4.1 Supply range. The distribution unit shall operate within specifications over a supply voltage range of 20 to 30 volts (V) dc. The dc power requirement shall not exceed 20 watts (W) at 26 V dc. The distribution unit shall not be damaged by operation from a supply operating at worst case limits the following conditions (see 4.5.4.1).

3.4.1.1 Steady-state voltage. Circuit steady-state voltage shall be between 25 and 30 V for combined generator-battery power supply, 20 to 27 V for battery only condition and no less than 33 V for generator only condition.

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3.4.1.2 Ripple. Upper and lower peaks of ripple voltage (see figure 3) shall each be less than 2 V for combined generator-battery power supply and battery only power condition and 7 V for generator only condition. The frequency components of the ripple shall be within the range of 50 Hertz (Hz) to 200 kilohertz (kHz).

3.4.1.3 Surges. All surges resulting from system operation shall fall within the loci shown in figure 4 for combined generator-battery power supply and in figure 5 for generator only condition. For battery only condition, any switching activity resulting in a surge which takes the voltage outside steady-state limits will be considered as a fault condition for the duration of the excursion.

3.4.1.4 Spikes. All spikes resulting from system operation shall fall within the loci shown in figure 6 for combined generator-battery power supply and battery only condition and in figure 5 for generator only condition.

3.4.2 Radio receive audio input impedance. The input impedance between pins A and K of J501, J503, J509 and J511 shall be 150 ohms $\pm$ 15 percent (%) each when measured at 1000 Hz (see 4.5.4.2).

3.4.3 Radio receive gain. The gain from pins A and K of J501 to pin J of J504, J505, J506 and J507 shall be 417 $\pm$ 23 millivolts (mV) per V when measured at 1000 Hz. The gain from pins A and K of J503 to pin M of J504, J505, J506 and J507; the gain from pins A and K of J509 to pin T of J504, J505, J506 and J507; and the gain from pins A and K of J511 to pin S of J504, J505, J506 and J507 shall also be 417 $\pm$ 23 mV per V when measured at 1000 Hz (see 4.5.4.3).

3.4.4 Radio receive frequency response. With a 0.22 V root mean square (rms) signal applied at the frequencies listed below, the output voltage shall be as specified below (see 4.5.4.4):

Frequency (Hz)	Output level (mV)
150	8 $\pm$ 3
300	80.5 $\pm$ 20
1000	91.7 $\pm$ 5
3500	97.9 $\pm$ 2.2
5000	98.3 $\pm$ 2.2

3.4.5 Radio receive distortion. The radio receive buffer shall perform as specified with a 0.22 V rms input at pin K of J501, J503, J509 and J511 with less than 2% total harmonic distortion (thd) when measured at 1000 Hz (see 4.5.4.5).

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3.4.6 Radio receive crosstalk. The output from any non-driven radio buffer shall be at least 50 decibels (dB) below the output of a driven radio buffer channel (see 4.5.4.6).

3.4.7 Radio receive buffer output impedance. The output impedance between pins J, M, T and S and pin A of J504, J505, J506 and J507 shall be 46 ohms ± 10% each when measured at 1000 Hz (see 4.5.4.7).

3.4.8 Intercom accent control. The intercom accent control shall reduce the output of the radio receive buffers 20 ± 3 dB when activated (see 4.5.4.8).

3.4.9 Microphone buffer audio input impedance. The input impedance between pins K, N, U and V and pin A of J504, J505, J506 and J507 shall be 68 ohms ± 10% each, when measured at 1000 Hz (see 4.5.4.9).

3.4.10 Microphone buffer gain. The voltage gain at 1000 Hz from pin K of J504, J505, J506 and J507 to pin U of J501; pin N of J504, J505, J506 and J507 to pin U of J503; pin U of J504, J505, J506 and J507 to pin U of J509; and pin V of J504, J505, J506 and J507 to pin U of J511 shall be 11.86 ± 0.8 V per V (see 4.5.4.10).

3.4.11 Microphone buffer frequency response. The microphone buffer frequency response shall be flat ± 2 dB from 300 to 3500 Hz (see 4.5.4.11).

3.4.12 Microphone buffer distortion. The microphone buffer shall perform as specified with a 35 mV, 1000 Hz signal applied at the input. The output shall contain less than 2% thd (see 4.5.4.12).

3.4.13 Microphone buffer crosstalk. The output from any non-driven microphone buffer shall be at least 50 dB below the output of a driven microphone buffer (see 4.5.4.13).

3.4.14 Microphone buffer output impedance. The output impedance between pins U and A of J501, J503, J509 and J511 shall be 150 ohms ± 10% each when measured at 1000 Hz (see 4.5.4.14).

3.4.15 Retransmit gain. With the retransmit switch on RETRANS, the voltage gain from pin K of J501 to pin U of J503 and from pin K of J503 to pin U of J501 shall be 2.01 ± 0.5 V per V when measured at 1000 Hz (see 4.5.4.15).

3.4.16 Line amplifier input impedance. The input impedance at the line terminals when measured at 1000 Hz shall be 600 ohms ± 10% balanced, when in the LISTEN mode (see 4.5.4.16).

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3.4.17 Line amplifier gain (listen mode). The voltage gain from the line terminals to pin R of J504, J505, J506 and J507 shall be 66.83 ± 3.06 mV per V each when measured at 1000 Hz (see 4.5.4.17).

3.4.18 Line amplifier distortion (listen mode). The line amplifier shall perform as specified with a 0.53 V rms signal applied to the input. The output shall contain less than 5% thd at 1000 Hz (see 4.5.4.18).

3.4.19 Line amplifier gain (talk mode). The voltage gain from pin R of J504, J505, J506 and J507 to the line terminals shall be 14.8 ± 1.96 V per V each when measured at 1000 Hz with the line terminals connected to a 600 ohm load (see 4.5.4.19).

3.4.20 Line amplifier distortion (talk mode). The line amplifier shall perform as specified with a 35 mV rms, 1000 Hz signal applied to the input. The output shall contain less than 10% thd (see 4.5.4.20).

3.4.21 Abnormal load (line amplifier). The line amplifier shall not be damaged by the application of 50 V rms, 20 Hz ringing signals while in the TALK mode or the LISTEN mode (see 4.5.4.21).

3.4.22 Auxiliary amplifier input impedance. The input impedance of the auxiliary terminals shall be 600 ohms ± 15% when measured at 1000 Hz (see 4.5.4.22).

3.4.23 Auxiliary amplifier gain. The voltage gain from the auxiliary input terminals to pin R of J504, J505, J506 and J507 shall be 4.79 ± 0.183 mV per V each when measured at 1000 Hz (see 4.5.4.23).

3.4.24 Auxiliary amplifier distortion. The auxiliary amplifier shall perform as specified with a 7 V rms, 1000 Hz signal applied to the input. The output shall contain less than 5% thd (see 4.5.4.24).

3.4.25 Abnormal input (auxiliary). The auxiliary amplifier shall not be damaged by the application of a 50 V rms, 20 Hz ringing signal (see 4.5.4.25).

3.4.26 Keying. Each distribution unit shall be capable of passing a keying signal (0.5 amp inductive load) to ground from pins D, E, F and H of J504 to pin S of J501, J503, J509 and J511, respectively. With pins D, E, F and H of J504 open circuited, the resistance from pin S to pin A on J501 shall exceed 10 000 ohms. With pins D, E, F and H of J504 grounded, the resistance between pins S and A on J501 (radio 1 selected) shall be less than 2 ohms. The keying circuit line shall be capable of withstanding an open circuit voltage equivalent to the power supply voltage specified in 3.4.1 (see 4.5.4.26).

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3.4.27 Environmental conditions. The distribution unit shall operate as specified in the environmental conditions specified herein. The distribution unit shall meet the performance requirements of this specification during exposure to designated environments while operating. The unit shall not be damaged nor its performance degraded after exposure to those environments designated nonoperating.

3.4.27.1 Temperature. The distribution unit shall operate in ambient air temperatures of minus (-) 32 to plus (+) 71 degrees Celsius (°C) [-25 to +160 degrees Fahrenheit (°F)]. The distribution unit shall withstand storage temperatures of -40 to +71°C (-40 to +160°F) for a duration of 6 hours (see 4.5.4.27.1).

3.4.27.2 Low pressure (altitude). The distribution unit shall operate at altitudes to 3000 meters (m) [10 000 feet (ft)]. Nonoperating, the distribution unit shall withstand exposure to altitudes to 12 000 m (40 000 ft) (see 4.5.4.27.2).

3.4.27.3 Humidity. The distribution unit shall operate in humidity conditions of 5 to 95% relative humidity (rh) (see 4.5.4.27.3).

3.4.27.4 Fungus. The distribution unit shall be operational after exposure to the fungi specified in MIL-STD-810, method 508.4, for 90 days at a rh of 95% and a temperature of 29°C (84°F) (see 4.5.4.27.4).

3.4.27.5 Salt fog. The distribution unit shall operate when exposed to salt fog conditions as defined in MIL-STD-810, method 509.3, procedure I (see 4.5.4.27.5).

3.4.27.6 Water spray. The performance of the distribution unit shall not be degraded (see 4.5.4.27.6).

3.4.27.7 Sand and dust. The distribution unit shall not be degraded by exposure to blowing sand or dust (see 4.5.4.27.7).

3.4.27.8 Explosive atmosphere. The distribution unit shall not cause ignition of an ambient explosive gaseous mixture of air when operating in such an atmosphere (see 4.5.4.27.8).

3.4.27.9 Vibration. The distribution unit shall pass the vibration requirement of method 514.4, category 8, procedure I of MIL-STD-810, with the test level as specified in figure 7 (see 4.5.4.27.9).

3.4.27.10 Shock, ballistic. The distribution unit shall withstand a shock test in accordance with ASTM D5487 (see 4.5.4.27.10).

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3.4.27.11 Electromagnetic interference. When specified (see 6.2), electromagnetic interference (EMI) suppression shall conform to the following (see 4.5.4.27.11).

3.4.27.11.1 Conducted emissions. Conducted emissions on power leads shall not exceed the values of 10 kHz to 10 millihertz (MHz), including returns that obtain power from other sources that are not part of the equipment under test (see figure 8).

3.4.27.11.2 Radiated emissions. Radiated emissions of the electrical field for Army systems shall not exceed the values of 2 MHz to 18 gigahertz (GHz). This requirement does not apply at the transmitter fundamental frequencies or to the radiated antennas. Above 30 MHz, the limits shall be met for both horizontal and vertically polarized fields (see figure 9).

3.5 Painting. Unless otherwise specified (see 6.2), surface treatments, coatings, and finishes for the distribution unit shall be in accordance with the best commercial practice standards. Finishes shall not be of the wrinkle type. Final color shall be a lusterless forest green (see 4.4.2, 4.5.2, and 6.3.2).

3.6 Identification and marking. The exterior markings shall be in accordance with ANSI/SAE AS478.

3.6.1 Safety markings. Unless otherwise specified (see 6.2), safety plates and decals shall be in accordance with applicable drawings (see 4.4.2 and 4.5.2).

3.6.2 Nameplate and product marking. Parts and assemblies shall be marked for identification in accordance with ANSI/SAE AS478. Whenever practicable, parts shall be so mounted that their identification marking will be readily visible. Nameplates shall be permanently attached to the equipment. Nameplates shall have the following information (see 4.4.2 and 4.5.2).

AMPLIFIER, AUDIO FREQUENCY
24 VDC
Part number _____
Serial number _____
National stock number _____
Manufacturer name and code _____
Identification number _____
Contract number _____
Date _____

3.7 Workmanship. Workmanship shall be such as to produce a finished product that is clean and free from defects that will affect its life, serviceability, or appearances. Workmanship shall be such as to ensure that the finished product is in conformance to this specification (see 4.4.2 and 4.5.2).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
   1. Preproduction inspection (see 4.3.1).
   2. Initial production inspection (see 4.3.2).
b. Conformance inspections (see 4.4).
   1. Acceptance tests (see 4.4.3.1).
   2. Control tests (see 4.4.3.2).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Air temperature: 23 ± 10°C (73 ± 18°F)
b. Barometric pressure: 725 (+50, -75) mm (28.5 (+2, -3) in.) Hg of mercury
c. Relative humidity: 50 ± 30%.

4.3 First article inspection. First article inspections (see 3.1) shall be performed on preproduction or initial production samples as specified herein.

4.3.1 Preproduction inspection. When specified (see 6.2), the preproduction sample shall consist of two distribution units. Preproduction inspection shall be as specified in table I.

4.3.2 Initial production inspection. Unless otherwise specified (see 6.2), two of the first ten distribution units produced under the production contract for initial production shall be selected. Initial production units shall be inspected as specified in table I.

4.3.2.1 In-process inspection. During fabrication of the first production distribution unit, in-process inspections will be conducted to evaluate conformance of materials and workmanship to requirements of specified documents. The distribution unit and all components shall be available for inspection. These inspections shall be made prior to application of primer and paint. Processing procedures, quality system, inspection records, and radiographic procedures will be reviewed and evaluated during the in-process inspections.

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TABLE I. Classification of inspection.

Title	Requirement	Inspection	First article	Conformance	
				Accept-ance	Con-trol
Materials and construction	3.2 thru 3.2.6, 3.3	4.5.1	X		X
Defects (see table II)	3.3, 3.3.1, 3.5 thru 3.7	4.5.2	X	X	
Size and weight	3.3.1	4.5.3	X		X
Interface description	3.3.2.1	4.5.3	X	X	
On-off switch	3.3.2.2	4.5.3	X	X	
Line jacks	3.3.2.3	4.5.3	X	X	
Auxiliary input	3.3.2.4	4.5.3	X	X	
Intercom cable assembly	3.3.2.5	4.5.3	X	X	
Power indicator	3.3.2.6	4.5.3	X	X	
Intercom accentuated switch	3.3.2.7	4.5.3	X	X	
Radio transmission selector switch	3.3.2.8	4.5.3	X	X	
Radio retransmission switch	3.3.2.9	4.5.3	X	X	
Line switch	3.3.2.10	4.5.3	X	X	
Power interface	3.3.2.11	4.5.3	X	X	
Power distribution	3.3.2.12	4.5.3	X	X	
Retransmission switch	3.3.2.13	4.5.3	X	X	
Supply range	3.4.1	4.5.4.1	X	X	
Radio receive audio input impedance	3.4.2	4.5.4.2	X	X	
Radio receive gain	3.4.3	4.5.4.3	X	X	
Radio receive frequency response	3.4.4	4.5.4.4	X	X	
Radio receive distortion	3.4.5	4.5.4.5	X	X	
Radio receive crosstalk	3.4.6	4.5.4.6	X	X	
Radio receive buffer output impedance	3.4.7	4.5.4.7	X	X	
Intercom accent control	3.4.8	4.5.4.8	X	X	
Microphone buffer audio input impedance	3.4.9	4.5.4.9	X	X	

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TABLE I. Classification of inspection - Continued.

Title	Requirement	Inspection	First article	Conformance	
				Accept-ance	Con-trol
Microphone buffer gain	3.4.10	4.5.4.10	X	X	
Microphone buffer frequency response	3.4.11	4.5.4.11	X	X	
Microphone buffer distortion	3.4.12	4.5.4.12	X	X	
Microphone buffer crosstalk	3.4.13	4.5.4.13	X	X	
Microphone buffer output impedance	3.4.14	4.5.4.14	X	X	
Retransmit gain	3.4.15	4.5.4.15	X	X	
Line amplifier input impedance	3.4.16	4.5.4.16	X	X	
Line amplifier gain (listen mode)	3.4.17	4.5.4.17	X	X	
Line amplifier distortion (listen mode)	3.4.18	4.5.4.18	X	X	
Line amplifier gain (talk mode)	3.4.19	4.5.4.19	X	X	
Line amplifier distortion (talk mode)	3.4.20	4.5.4.20	X	X	
Abnormal load (line amplifier)	3.4.21	4.5.4.21	X	X	
Auxiliary amplifier input impedance	3.4.22	4.5.4.22	X	X	
Auxiliary amplifier gain	3.4.23	4.5.4.23	X	X	
Auxiliary amplifier distortion	3.4.24	4.5.4.24	X	X	
Abnormal input (auxiliary)	3.4.25	4.5.4.25	X	X	
Keying	3.4.26	4.5.4.26	X	X	
Temperature	3.4.27.1	4.5.4.27.1	X		
Low pressure (altitude)	3.4.27.2	4.5.4.27.2	X		
Humidity	3.4.27.3	4.5.4.27.3	X		
Fungus	3.4.27.4	4.5.4.27.4	X		
Salt fog	3.4.27.5	4.5.4.27.5	X		
Water spray	3.4.27.6	4.5.4.27.6	X		
Sand and dust	3.4.27.7	4.5.4.27.7	X		

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TABLE I. Classification of inspection - Continued.

Title	Requirement	Inspection	First article	Conformance	
				Accept-ance	Con-trol
Explosive atmosphere	3.4.27.8	4.5.4.27.8	X		
Vibration	3.4.27.9	4.5.4.27.9	X		
Shock, ballistic	3.4.27.10	4.5.4.27.10	X		
Electromagnetic interference	3.4.27.11	4.5.4.27.11	X		

4.4 Conformance inspection.

4.4.1 Sampling.

4.4.1.1 Lot formation. An inspection lot shall consist of all the distribution units of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for examination. Samples from an inspection lot for conformance examination shall be selected in accordance with ANSI/ASQC Z1.4.

4.4.2 Conformance examinations.

4.4.2.1 Lot acceptance. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.

4.4.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

4.4.3 Conformance tests.

4.4.3.1 Acceptance tests. Each distribution unit shall be subjected to the tests specified in table I.

4.4.3.2 Control tests. Control tests shall be conducted on one distribution unit from each lot of 250 units consecutively produced, except that not more than one test shall be performed in a three-month period, nor less than one test in a six-month period. The distribution unit shall be subjected to the tests specified in table I.

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TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	Visual
Major:		
101	Assembly, incomplete (see 3.3, 3.3.2.2 thru 3.3.2.13).	
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIE 1/
103	Electrical interface, improper (see 3.3.3).	Visual
104	Identification marking, missing, incomplete, or not legible (see 3.6 thru 3.6.2).	Visual
105	Faulty workmanship affecting performance (see 3.7).	Visual
Minor:		
201	Wiring and cables, defective, missing, improper installation, coding or protection (see 3.3).	Visual
202	Fasteners, improper assembly or torque (see 3.3).	Visual and SIE
203	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
204	Paint, improper application or color (see 3.5).	Visual
205	Identification marking, improper location or size, defective (see 3.6 thru 3.6.2).	Visual
206	Faulty workmanship affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3.3 Failure. Failure of any distribution unit to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5 Methods of inspection.

4.5.1 Materials and construction. Conformance to 3.2 through 3.2.6, and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.5.2 Defects. Conformance to 3.3, 3.3.1 and 3.5 through 3.7 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.5.3 Features. Conformance to 3.3.1, 3.3.2.1 through 3.3.2.13, and 3.4.1 shall be determined by exercising the distribution unit, and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

4.5.4 Performance tests. Except as otherwise specified herein, equipment performance shall be measured with a 333 ohm resistor in parallel with an 825 ohm -0.27 microfarad series combination connected between pin R and signal ground in order to simulate system operation. Except as otherwise specified herein, gain measurements shall be made with equipment volume controls set to maximum.

4.5.4.1 Supply range. To determine conformance to 3.4.1, the equipment shall be operated from a 26 V dc power supply and the current shall be measured and shall not exceed 0.77 amperes. The equipment shall be operated from a supply operating at the worst case limits (see 3.4.1). The supply need not exhibit the time requirements (see 3.4.1.2, 3.4.1.3 and 3.4.1.4). However, the applicable curves shown (see figures 3, 4, 5, and 6) shall be contained with the resulting voltage-time characteristic of the power supply used. The equipment shall not be damaged or degraded.

4.5.4.2 Radio receive audio input impedance. To determine conformance to 3.4.2, the input impedance shall be measured between pins A and K of J501, J503, J509 and J511 at 1000 Hz.

4.5.4.3 Radio receive gain. To determine conformance to 3.4.3, a 0.22 V rms, 1000 Hz signal shall be applied to pins A and K of J501. With the output taken at pin J of J504, J505, J506 and J507, each output shall measure 91.7 $\pm$ 5.0 mV into an open circuit. The same signal shall be applied to pins A and K of J503, J509 and J511 and the output taken at pin M of J504, J505, J506 and J507; pin T of J504, J505, J506 and J507; and pin S of J504, J505, J506 and J507, respectively. Each output shall measure 91.7 $\pm$ 5.0 mV into an open circuit. This circuit corresponds to a gain of 416 $\pm$ 23 mV per V.

4.5.4.4 Radio receive frequency response. To determine conformance to 3.4.4, a constant amplitude variable frequency signal shall be applied. With the output taken as specified in 4.5.4.3, the output voltage shall be as specified in 3.4.4.

4.5.4.5 Radio receive distortion. To determine conformance to 3.4.5, a 0.22 V rms, 1000 Hz signal shall be applied to pin K of J501. With the output taken at pin J of J504, J505, J506 and J507, the distortion shall be less than 2% at each output. The same signal shall be

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applied to pin K of J503, J509 and J511 and the output taken at pin M of J504, J505, J506 and J507; pin T of J504, J505, J506 and J507; and pin S of J504, J505, J506 and J507, respectively. The distortion shall be less than 2% at each output.

4.5.4.6 Radio receive crosstalk. Under the conditions specified in 4.5.4.5, the output from any non-driven channel shall be at least 50 dB below the output of a driven channel to determine conformance to 3.4.6.

4.5.4.7 Radio receive buffer output impedance. To determine conformance to 3.4.7, the output impedance shall be measured between pins J, M, T and S and pin A of J504, J505, J506 and J507 at 1000 Hz.

4.5.4.8 Intercom accent control. To determine conformance to 3.4.8, the test specified in 4.5.4.3 shall be performed except the intercom accent switch shall be set to "ON". Output power shall be 20 ± 3 dB below the level observed in 4.5.4.3.

4.5.4.9 Microphone buffer audio input impedance. To determine conformance to 3.4.9, the input impedance shall be measured between pins, K, N, U and V and pin A of J504, J505, J506 and J507 at 1000 Hz.

4.5.4.10 Microphone buffer gain. To determine conformance to 3.4.10, a 35 mV signal shall be applied to pin K of J504, J505, J506 and J507. With the output taken at pin U of J501, each output shall measure 415 ± 28 mV into an open circuit at 1000 Hz. The same signal shall be applied to pin N of J504, J505, J506 and J507; pin U of J504, J505, J506 and J507; pin V of J504, J505, J506 and J507; and output taken at pin U of J503, J509 and J511, respectively. Each output shall measure 415 ± 28 mV into an open circuit at 1000 Hz. This output corresponds to a gain of 11.86 ± 0.8 V per V.

4.5.4.11 Microphone buffer frequency response. To determine conformance to 3.4.11, a constant amplitude variable frequency signal shall be applied. With the output taken as specified in 4.5.4.10, the output voltage shall be as specified below:

Frequency (Hz)	Output level (dB)
300	± 2
1000	0 (reference 415 ± 28 mV)
3500	± 2

4.5.4.12 Microphone buffer distortion. To determine conformance to 3.4.12, a 35 mV, 1000 Hz signal shall be applied to pin K of J504, J505, J506 and J507. With the output taken at pin U of J501, the distortion shall be less than 2%. The same signal shall be applied to pin N of

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J504, J505, J506 and J507; pin U of J504, J505, J506 and J507; pin V of J504, J505, J506 and J507; and output taken at pin U of J503, J509 and J511, respectively. The distortion shall be less than 2%.

4.5.4.13 Microphone buffer crosstalk. Under the conditions specified in 4.5.4.12, the output from any non-driven channel shall be at least 50 dB below the output from a driven channel to determine conformance to 3.4.13.

4.5.4.14 Microphone buffer output impedance. To determine conformance to 3.4.14, the output impedance shall be measured between pins U and A of J501, J503, J509 and J511 at 1000 Hz.

4.5.4.15 Retransmit gain. To determine conformance to 3.4.15, a 0.22 V rms, 1000 Hz signal shall be applied to pin K of J501 and pin K of J503 with the retransmit switch on RETRANS. With the output taken at pin U of J503 and pin U of J501 respectively, each output shall measure 442 $\pm$110 mV into an open circuit. This output corresponds to a gain of 2.01 $\pm$ 0.5 V per V.

4.5.4.16 Line amplifier input impedance. To determine conformance to 3.4.16, the input impedance shall be measured at the line terminals at 1000 Hz, when in the LISTEN mode.

4.5.4.17 Line amplifier gain (listen mode). To determine conformance to 3.4.17, a 0.53 V rms, 1000 Hz signal shall be applied to the line terminals. With the output taken at pin R of J504, J505, J506 and J507, the output voltage shall measure 35.4 $\pm$ 1.6 mV. This output corresponds to a gain of 66.83 $\pm$ 3.06 mV per V.

4.5.4.18 Line amplifier distortion (listen mode). To determine conformance to 3.4.18, a 0.53 V rms, 1000 Hz signal shall be applied to the line terminals. With the output taken at pin R of J504, J505, J506 and J507, the distortion shall not exceed 5%.

4.5.4.19 Line amplifier gain (talk mode). To determine conformance to 3.4.19, a 35 mV rms, 1000 Hz signal shall be applied to pin R of J504, J505, J506 and J507. With the output taken at the line terminals, the output voltage shall measure 518 $\pm$ 68.6 mV each with the output terminated into 600 ohms. This output corresponds to a gain of 14.8 $\pm$ 1.96 V per V.

4.5.4.20 Line amplifier distortion (talk mode). To determine conformance to 3.4.20, a 35 mV rms, 1000 Hz signal shall be applied to pin R of J504, J505, J506 and J507. With the output taken at the line terminals terminated into 600 ohms, the distortion shall not exceed 10%.

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4.5.4.21 Abnormal load (line amplifier). To determine conformance to 3.4.21, the test specified in 4.5.4.20 shall be performed. A 50 V rms, 20 Hz signal shall be applied across the line terminals. The tests specified in 4.5.4.17 through 4.5.4.20 shall be repeated. No damage or degradation shall occur.

4.5.4.22 Auxiliary amplifier input impedance. To determine conformance to 3.4.22, the input impedance of the auxiliary terminals shall be measured at 1000 Hz.

4.5.4.23 Auxiliary amplifier gain. To determine conformance to 3.4.23, a 7 V rms, 1000 Hz signal shall be applied to the auxiliary input terminals. With the output taken at pin R of J504, J505, J506 and J507, each output shall measure 33.5 $\pm$ 1.3 mV. This output corresponds to a gain of 4.79 $\pm$ 0.183 mV per V.

4.5.4.24 Auxiliary amplifier distortion. To determine conformance to 3.4.24, an input signal of 7 V rms, 1000 Hz shall be applied to the auxiliary input terminals. With the output at pin R of J504, J505, J506 and J507, the distortion shall not exceed 5%.

4.5.4.25 Abnormal input (auxiliary). To determine conformance to 3.4.25, the tests specified in 4.5.4.23 and 4.5.4.24 shall be performed, except that a 50 V rms, 20 Hz signal shall be applied to the auxiliary input terminals. No damage or degradation shall occur.

4.5.4.26 Keying. To determine conformance to 3.4.26, a keying signal (0.5 amp inductive load) shall be passed to ground from pins D, E, F and H of J504 to pin S of J501, J503, J509 and J511, respectively. With pins D, E, F and H of J504 open circuited, the resistance from pin S to pin A on J501 shall exceed 10 000 ohms. With pins D, E, F and H of J504 grounded, the resistance between pins S and A on J501 (radio 1 selected) shall be less than 2 ohms. The keying circuit shall be opened and the keying line subjected to power supply voltages per 4.5.4.1 via an inductive-resistive device (relay coil or equivalent) resistance of 56 ohms $\pm$ 10% and inductance of 10 milliHenry $\pm$ 20%. The keying circuit shall be exercised and no degradation shall occur.

4.5.4.27 Environmental tests.

4.5.4.27.1 Temperature. To determine conformance to 3.4.27.1, the distribution unit shall be tested as follows:

a. Operation.

1. Low temperature. The distribution unit shall be tested for low temperature operation by first storing it for 6 hours in a test chamber at -32°C (-25°F), then, with the unit still held to -32°C, performing the tests of 4.5.4.1 through 4.5.4.26.

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2. High temperature. The distribution unit shall be tested for high temperature by repeating the procedure used in the low temperature test except that the temperature of the test chamber and the distribution unit shall be 71°C (160°F).

b. Storage.

1. Low temperature. The distribution unit shall be tested for low temperature storage by first storing the unit for 6 hours in a test chamber at -40°C (-40°F). Then, with the unit in the chamber, the chamber temperature shall be increased to 23 ± 3°C (73 ± 5°F) and maintained for 6 hours. The tests of 4.5.4.1 through 4.5.4.26 shall be performed with the chamber temperature at 23 ± 3°C.
2. High temperature. The distribution unit shall be tested for high temperature storage by first storing the unit for 6 hours in a test chamber at 71°C (160°F). Then, with the unit in the chamber, the chamber temperature shall be increased to 23 ± 3°C, and maintained for 6 hours. The tests of 4.5.4.1 through 4.5.4.26 shall be performed with the chamber temperature at 23 ± 3°C.

4.5.4.27.2 Low pressure (altitude). To determine conformance to 3.4.27.2, the distribution unit shall be tested in accordance with MIL-STD-810, method 500.3 as follows:

a. Procedure I. The air pressure shall be equivalent to an altitude of 12 000 m (40 000 ft).
b. Procedure II. The air pressure shall be equivalent to an altitude of 3000 m (10 000 ft).

4.5.4.27.3 Humidity. To determine conformance to 3.4.27.3, the distribution unit shall be tested in accordance with MIL-STD-810, method 507.3, procedure II. The conditions shall conform to figure 507.3-2, cycle 5, except the humidity limits shall be 5 and 95% and the number of test cycles shall be ten. Equipment shall not be moved from the humidity chamber for measurements or power left on after measurements have been completed. If any unit fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement or parts shall be permitted. If the unit still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the unit fails to meet applicable requirements after final conditioning and adjustment, it does not pass the test. The test shall be performed with the cover off.

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4.5.4.27.4 Fungus. To determine conformance to 3.4.27.4, the distribution unit shall be subjected to 90 days of the fungus test specified in MIL-STD-810, method 508.4. The parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall be sprayed. The distribution unit shall be subjected to the test in the open condition. The distribution unit shall show no evidence of viable fungus or corrosion.

4.5.4.27.5 Salt fog. To determine conformance to 3.4.27.5, the distribution unit shall be tested in accordance with MIL-STD-810, method 509.3, procedure I.

4.5.4.27.6 Water spray. To determine conformance to 3.4.27.6, the distribution unit shall be sprayed for 5 minutes with a solid stream of water at a rate of 65 gallon per minute (gal/min) through a 1 in. diameter nozzle, when mounted by its normal mounting means.

4.5.4.27.7 Sand and dust. To determine conformance to 3.4.27.7, the distribution unit shall be tested in accordance with MIL-STD-810, method 510.3, procedures I and II, except the temperature during test step 5 shall be limited to 63°C (145°F) and the sand particles shall be 0.01 to 1.00 mm (0.39 to 39 mils) in diameter with predominant diameters between 0.15 and 0.30 mm (5.9 to 11.8 mils), and with winds greater than 55.6 kilometers per hour (km/hr) (30 knots). The dust shall have a density of 0.006 micrograms per cubic centimeter ($\mu g/cm^3$) [0.0026 grain per cubic foot (grain/ft^3)], with particle size 0.0001 to 0.01 mm (0.0039 to 0.39 mils) in diameter, and winds at 64.8 km/hr (35 knots).

4.5.4.27.8 Explosive atmosphere. To determine conformance to 3.4.27.8, the distribution unit shall be tested in accordance with MIL-STD-810, method 511.3, procedure I.

4.5.4.27.9 Vibration. To determine conformance to 3.4.27.9, the distribution unit shall be tested in accordance with MIL-STD-810, method 514.4, category 8, procedure I, with the test level as specified in figure 3. The distribution unit shall be operated during the test by injecting a 1000 Hz tone into all radio input channels. The output power shall be monitored and shall remain constant. The output tone shall be monitored and there shall be no repetitive break up or garbling. Channels shall be periodically reselected. A sinusoidal cycling time of 30 minutes per 1609 kilometers (km) (1000 miles) or a maximum of 3 hours is required. Sweep time is 15 minutes from 5 to 500 to 5 Hz.

4.5.4.27.10 Shock, ballistic. To determine conformance to 3.4.27.10, the distribution unit shall be tested in accordance with ASTM D5487. A distribution unit mounted on a bracket shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the unit. The test shall, through appropriate orientation of the unit under test, deliver a total of 9 blows: One each 1 foot blow, 3 foot blow, and 5 foot blow on the back, side, and top of the unit. Upon completion of the tests, and before repairs are made, the unit shall show no degradation of performance, damage to mounting hardware, or failure to interface with associated equipment.

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4.5.4.27.11 EMI. To determine conformance to 3.4.27.11, the EMI shall be tested in accordance with the applicable provisions of MIL-STD-462.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The distribution unit is intended to be used in, but not limited to, the M981 fire support team vehicle (FISTV).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Title, number, and date of applicable drawings.
d. When first article samples are required (see 3.1).
e. If EMI suppression is required (see 3.4.27.11).
f. If surface treatments, coatings, and finishes are other than specified (see 3.5).
g. Information to be included for identification and marking (see 3.6).
h. If safety markings should be other than as specified (see 3.6.1).
i. If inspection conditions should be other than as specified (see 4.2).
j. If preproduction inspection is required (see 4.3.1).
k. If initial production inspection is not required (see 4.3.2).
l. Packaging requirements (see 5.1).

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6.3 Definitions.

6.3.1 Item definition. The distribution unit is the main junction box between the intercommunication control units and the radio system installed in the vehicle, providing two-way communication between crew members and external personnel. In addition, a retransmission switch on the distribution unit enables two radios to be used alternately as a radio relay.

6.3.2 Forest green. Federal standard FED-STD-595 defines forest green using color chip number 34079.

6.4 Subject term (key word) listing.

Communication
Distribution unit
Fire support team vehicle
Gain
Main junction box

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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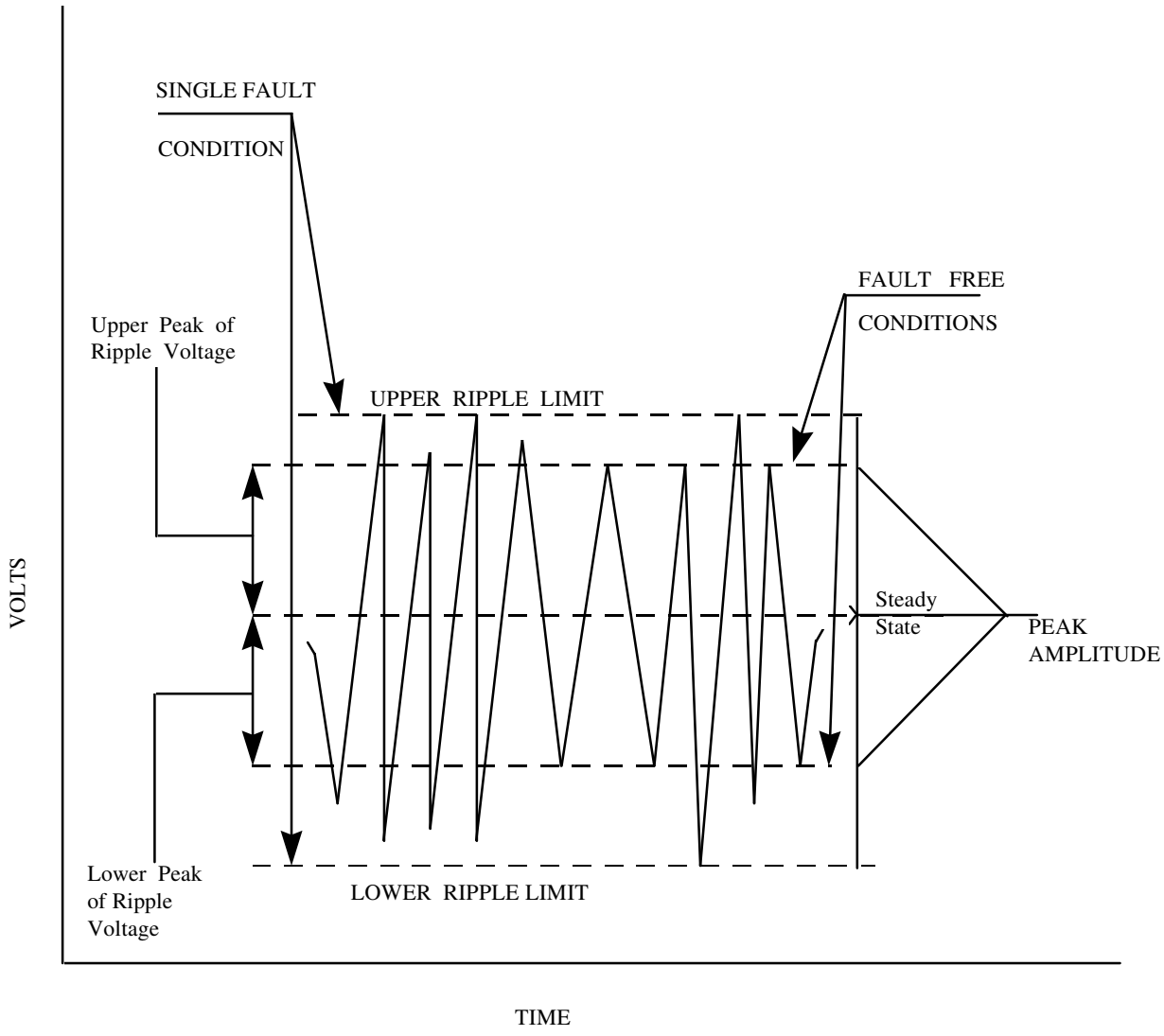


FIGURE 3. Enlarged view of ripple.

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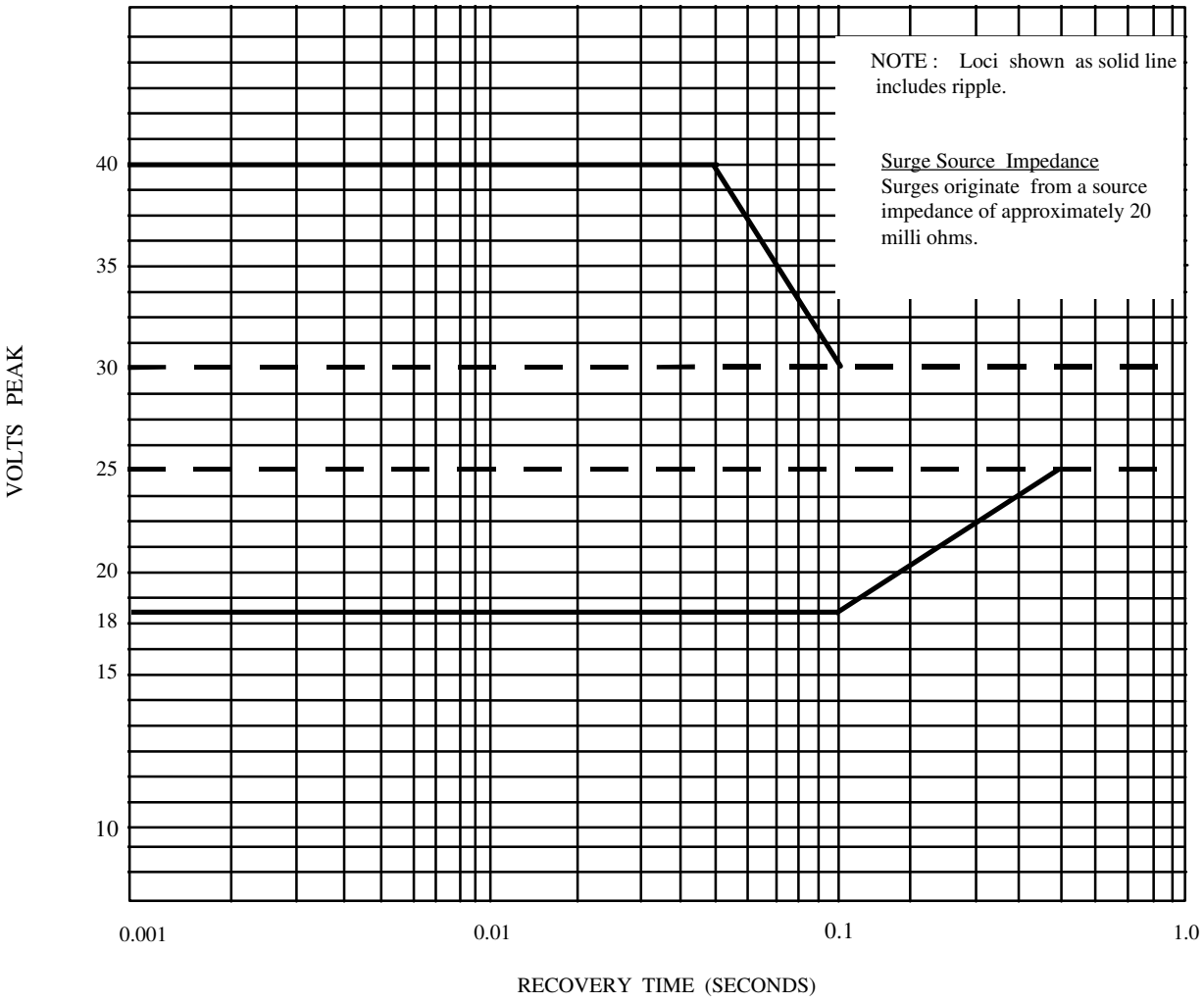


FIGURE 4. Loci of surges fault free condition.

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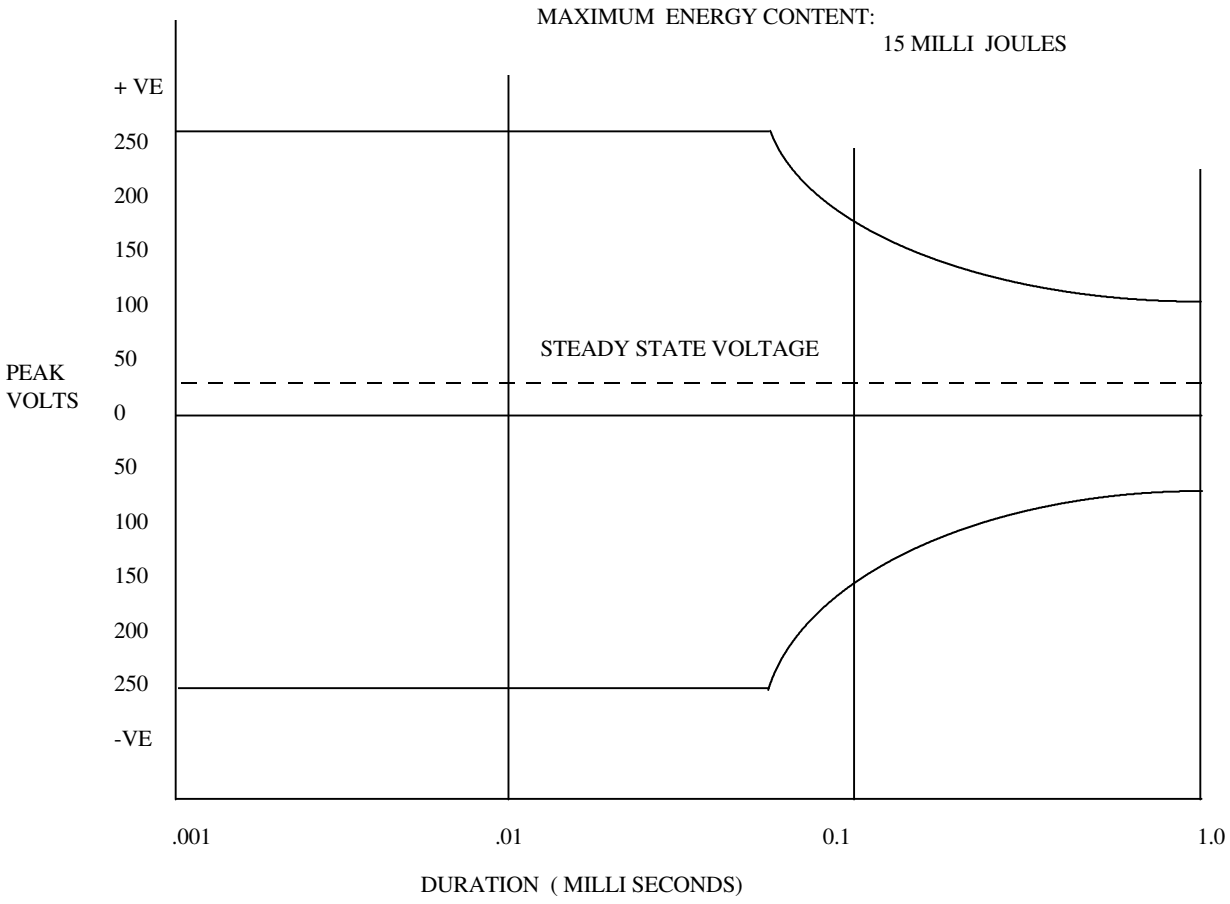


FIGURE 5. Loci of surges single fault condition.

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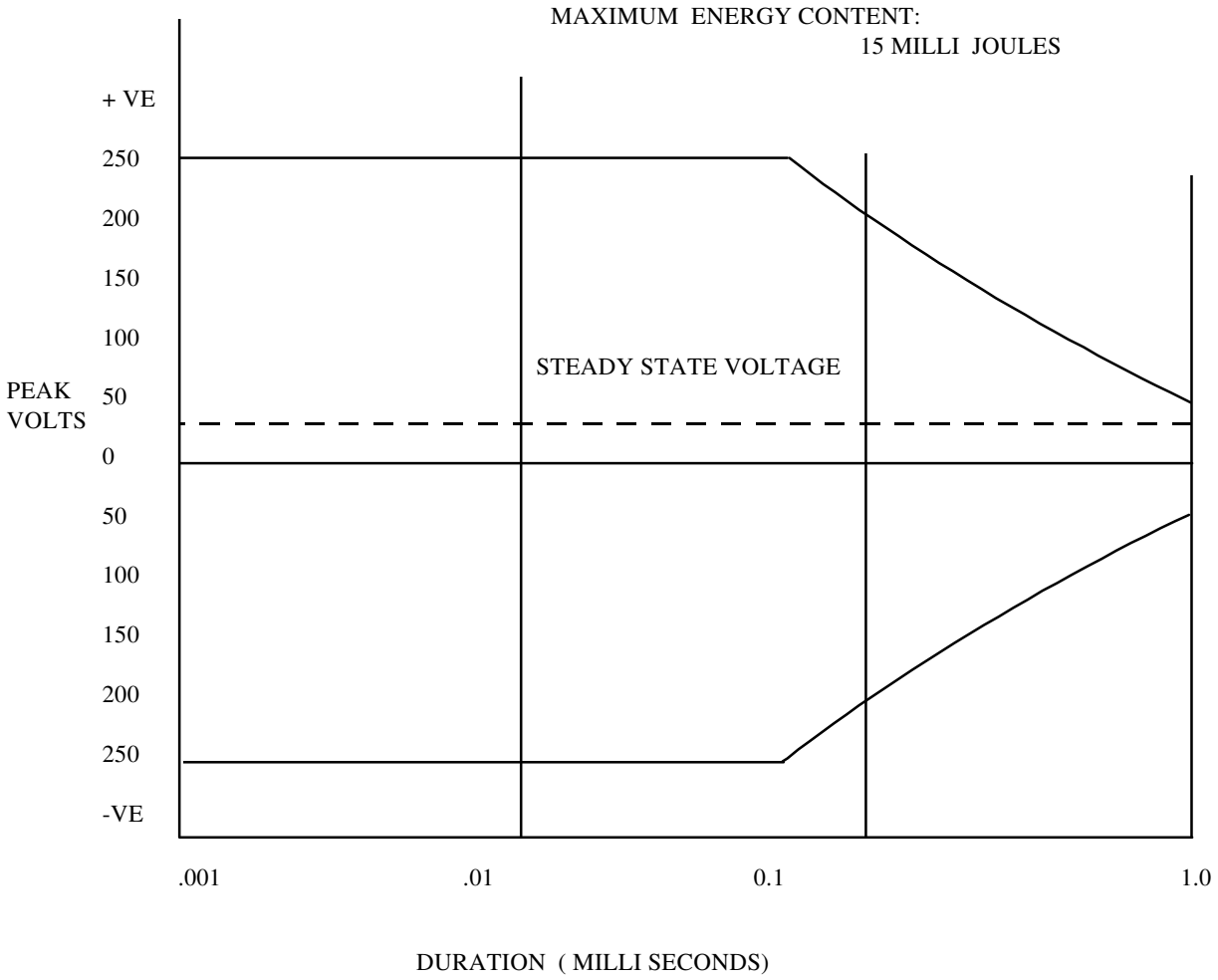
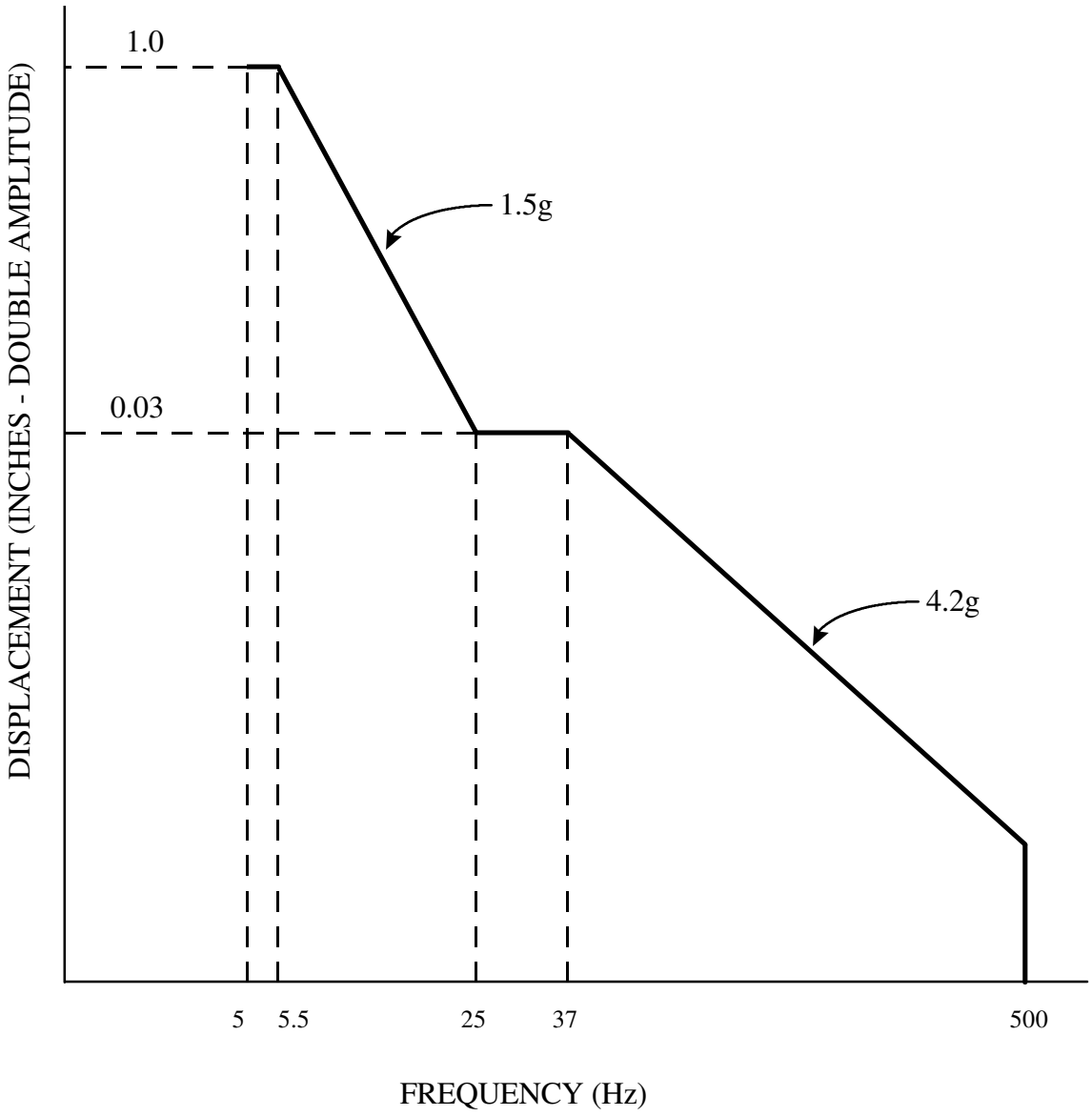


FIGURE 6. Loci of spikes fault free condition.

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NOTES: 1. All curves shall be extended to 2 Hz when test item resonances below 5 Hz are expected.
2. g = gravity unit.

FIGURE 7. Vibration test curve.

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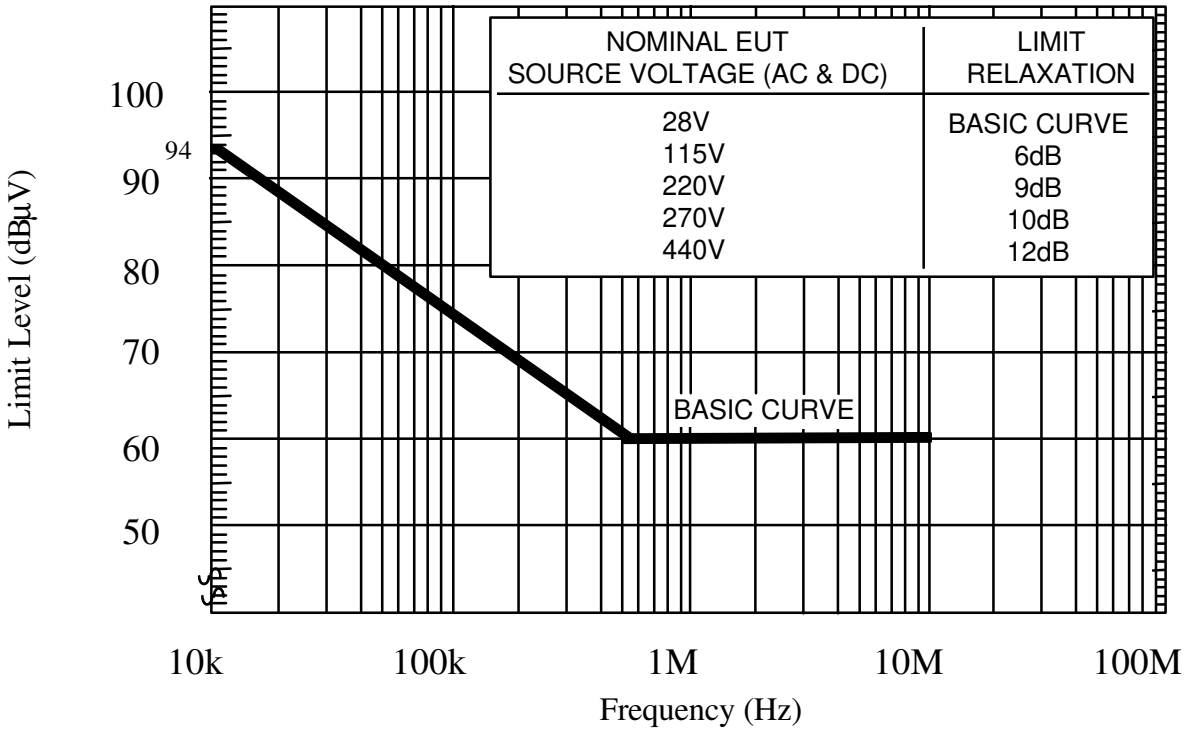


FIGURE 8. Conducted emissions limit (EUT power leads, AC and DC) for all applications.

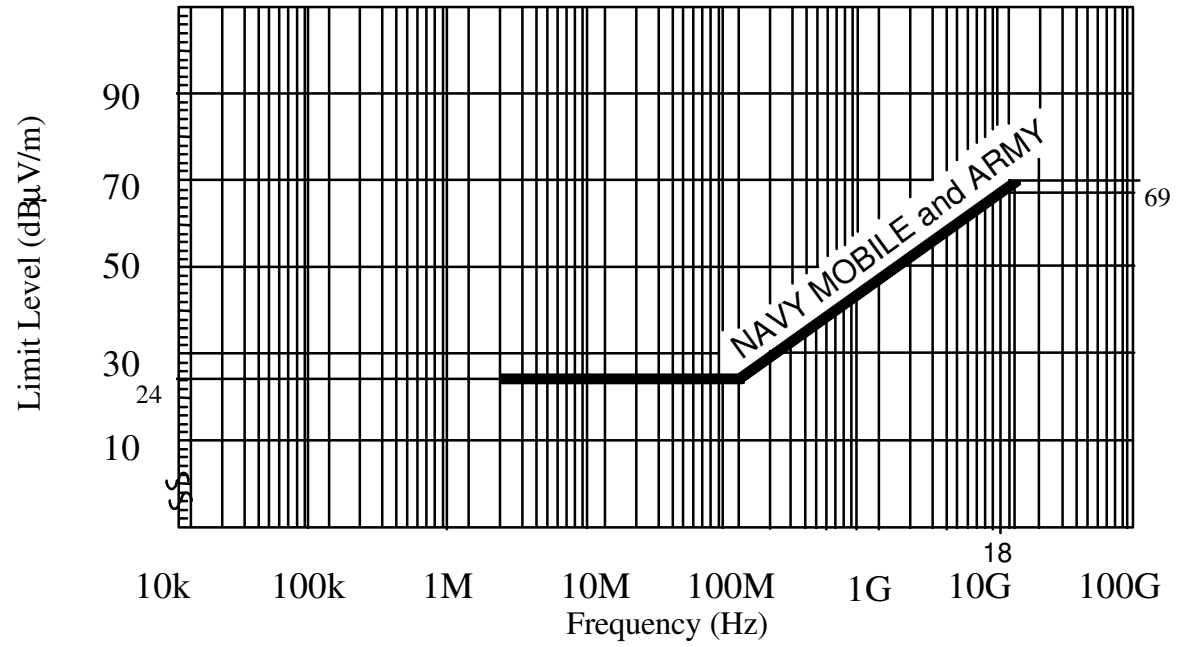


FIGURE 9. Radiated emissions limit for ground applications.

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AMPLIFIER, AUDIO FREQUENCY

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